

Name: White, Rice, & Spice Team**Date:** 3/9/26**Class:** Design for Engineers e4usa

MyDesign Portfolio

Element B: Documentation and Analysis of Prior Design Attempts

Prior Design Attempt #1 - Automatic Soap Dispenser

https://www.reliablepaper.com/product/Touch-Free-Foam-Soap-Dispenser-1200-mL-6-x-405-x-1058-Black__BWK42200.aspx



Details

The automatic soap dispenser design is commonly found across public spaces, but is not often seen in the home because of its high cost. Its touchless appeal is beneficial, and the design has been mass-produced.

Analysis

This design's primary focus is to dispense soap without touch, not to prevent soap puddles from forming, which prevents it from being valuable to stakeholders. Additionally, these costly machines can break down easily, are not accessible to the everyday home, further limiting the scope of this design to a different problem.

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Prior Design Attempt #2 - Paper Towels

<https://shop.bountytowels.com/bounty-regular-roll-white-select-a-size/>



Details

The paper towel is a home staple that has been used to clean up a variety of spills, including soap puddles that form after handwashing. These disposable towels are highly absorbent and, combined with user wiping, allow someone to clean a surface. Additionally, this item has been mass-produced. This design is separate from a hand towel as they are disposable and reusable but are more convenient for the user.

Analysis

The paper towel only offers a temporary solution to the problem: Nothing is stopping the soap puddles from forming again and again if the user doesn't wipe down the sink surface. One major problem stakeholders have with paper towels is that they fail to wipe around the soap bottle, which will guarantee some spots of soap puddles to be left. Finally, the design is unsustainable for the environment as it can lead to wasted paper.

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Prior Design Attempt #3 - Hand Towels

<https://cottoncreations.com/product/spa-comfort-hand-towel/>



Details

This common hand towel is used in many homes all over the world and has been used for hundreds of years. Traditionally, the design is reusable, uses wiping motions, and comes in a variety of materials. This design is separate from the paper towels because it is reusable and washable.

Analysis

Disadvantages that block the hand towel from being a feasible solution include the need to dry and the inaccessibility of cleaning. For example, with large puddles, the towel becomes ineffective, as the towel will need to dry before cleaning. Additionally, the hand towel lacks the ability to reach every area of the soap puddle, as the soap dispenser will block wiping attempts.

Prior Design Attempt #4 - Mat Design

<https://www.amazon.com/Folding-Faucet-Kitchen-Self-draining-Draining/dp/B0DR17XWZK?th=1>



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Details

This product and its variations' goal is to guide water that would've gone on the sink to the inside of the sink. The silicon design requires vast amounts of space and uses gravity as a potential solution.

Analysis

Major problems of the mat designs include the lack of proper sizes, as the mat is frequently too large or too small for most sinks. The design is very inaccessible to wiping and cleaning, leading to bacteria and mold growth. For these reasons, the mat design becomes an ineffective way of preventing soap puddles.